

Math 107-01
Final Exam A: December 13, 2013

Answer all of the following questions. You will have 2 hours to complete the exam. You may not use any outside assistance on this exam. The use of electronic equipment such as mp3 players, ipods, cell phones and any electronic devices other than your calculator during the exam is prohibited. If you have time, check your answers for accuracy.

Be sure to show all work. Unsupported answers may not receive credit.

Good luck!

Name: _____

1. Let h represent the statement "We do our homework."
Let m represent the statement "We miss classes."
Let w represent the statement "We do well in the class."
Using these letters symbolize the following statements (4 points each):
 - (a) We do our homework or we miss classes.
 - (b) If we do our homework and we do not miss any classes, then we should do well in the class.

2. Consider the original statement: "If the weather is nice, then we do not have to stay indoors." After reading statements in a,b,c, and d, you will be asked to identify the converse, inverse, and contrapositive of the original statement (4 points each). Write the letter of your answer choice on the line provided:
 - (a) If the weather is not nice, then we will have to stay indoors.
 - (b) If we have to stay indoors, then the weather is not nice.
 - (c) We will not have to stay indoors if the weather is nice.
 - (d) If we do not have to stay indoors, then the weather is nice.

Which statement is the converse of the original? _____

Which statement is the inverse of the original? _____

Which statement is the contrapositive of the original? _____

3. Construct a truth table for the following statement: $(p \vee \sim q) \rightarrow r$ (9 points)

p	q	r			
T	T	T			
T	T	F			
T	F	T			
T	F	F			
F	T	T			
F	T	F			
F	F	T			
F	F	F			

For problems 4-8, identify the negation of each statement (4 points each). Write the letter of your answer choice on the line provided:

4. No students may park in the faculty lot.
- a) Some students may park in the faculty lot.
 - b) Some students may not park in the faculty lot.
 - c) All students may park in the faculty lot.
 - d) None of the above.

Negation: _____

5. Some students are on the basketball team.
- a) All students are on the basketball team.
 - b) Some students are not on the basketball team.
 - c) No students are on the basketball team.
 - d) None of the above.

Negation: _____

6. All students enjoy Math 107.
- a) No students enjoy Math 107.
 - b) Some students enjoy Math 107.
 - c) Some students do not enjoy Math 107.
 - d) None of the above.

Negation: _____

7. If we do not pass the test, then we must take it again.
- a) We do not pass the test and we do not take it again.
 - b) We pass the test and we do not take it again.
 - c) If we do not pass the test, then we must take it again.
 - d) If we pass the test, then we must not take it again.

Negation: _____

8. We must buy the tickets or we will not be able to attend the concert.
- a) We do not buy the tickets and we are able to attend the concert.
 - b) We do not buy the tickets and we are not able to attend the concert.
 - c) We must buy the tickets and we are not able to attend the concert.
 - d) We must buy the tickets and we are able to attend the concert.

Negation: _____

9. Use a truth table to determine if the following argument is valid. (7 points)

$$\begin{array}{l} 1. q \vee r \\ 2. r \\ \hline \therefore \sim q \end{array}$$

		1	2		c	Argument
q	r					
T	T					
T	F					
F	T					
F	F					

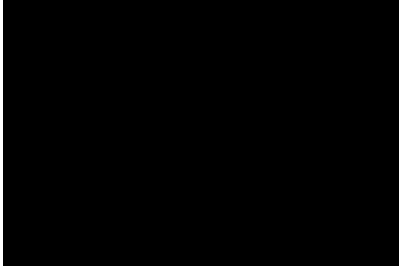
Valid or Invalid? _____

10. Use a Venn diagram to determine if the following argument is valid. (6 points)

1. All basketball players are tall.

2. Jack is a basketball player.

Therefore, Jack is tall.



Valid or Invalid? _____

11. Let $U = \{7, 14, 21, 28, 35, 42, 49, 56\}$ be the universal set for the problem. Let A and B be subsets of U with $A = \{7, 14, 21, 28\}$ and $B = \{14, 28, 42, 56\}$. Find the following (2 points each):

(a) $A' =$

(b) $B' =$

(c) $A \cap B =$

(d) $A \cup B =$

(e) $A' \cap B =$

(f) $(A \cup B)' =$

12. The theater club has 38 members. 20 members are taking an English course this semester. 17 of the members are taking a music course this semester. 6 of the club

members taking a music course are NOT taking an English course this semester. Let M represent the set of members taking a music course and E the set of members taking an English course. Draw a Venn diagram and label the value in each region (4 points).



13. There are 50 students in one residence hall. Before the students receive their grades for the end of the semester, they must pay all fines owed to the school. 29 of the students have an overdue library book fine (L). 37 of the students have a parking ticket (P) that must be paid. 20 of the students have an overdue library book fine and a parking ticket.



- (a) How many of the students have an overdue library book fine or a parking ticket? (4 points)
- (b) How many of the students have a parking ticket but not an overdue library book fine? (4 points)

14. Evaluate the following (3 points each):

(a) $7!$

(b) ${}_5P_3$

(c) ${}_9C_5$

15. A parking decal at a college consists of 2 letters followed by 5 digits. Each letter and digit may be used only once (in other words, letters and digits cannot be repeated). How many parking decals can be created using this method? (5 points)

16. An events committee of 7 students will be selected from a group of 11 freshmen and 24 upperclassmen. How many committees are possible if the committee must have 3 freshmen and 4 upperclassmen? (6 points)

17. A card is drawn from a well-shuffled deck. State all probabilities as reduced fractions (3 points each)

(a) How many outcomes are in the sample space?

(b) What is the probability of drawing a jack?

(c) What is the probability of drawing a spade?

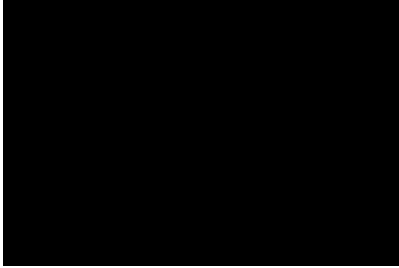
(d) What is the probability of drawing the jack of spades?

(e) What is the probability of drawing a jack or a spade?

(f) What are the odds of drawing a heart?

18. A player draws 6 cards from a well shuffled-deck. What is the probability that exactly two of these cards are fours? (5 points) Round your answer to 4 decimal places.

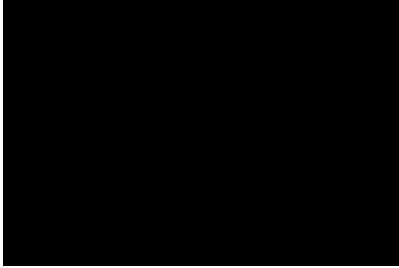
19. In LAA 121, 75% of the students are taking an English course and 60% of the students are taking a History course. 50% of the students are taking both English and History courses. Let E = “taking an English course” and H = “taking a History course”. The Venn diagram is given below.



Calculate the following probabilities (3 points each):

- (a) A student is taking an English course and is not taking a History course.
 - (b) A student is taking an English course or a History course.
 - (c) A student is taking a History course, given that he is taking an English course.
 - (d) A student is taking an English course, given that he is taking a History course.
20. (4 points) Let A and B be events with $p(A) = 0.35$, $p(B) = 0.20$ and $p(A \cap B) = 0.070$. Are A and B independent events? Why?

21. Greta's Grocery gets 60% of their apples from Olivia's Orchard and the rest from Fred's Farm. 4% of the apples from Olivia's Orchard are rotten and 2% of the apples from Fred's Farm are rotten. Draw a probability tree representing this, including the probabilities of the limbs (10 points)



22. The frequency distribution below shows the hourly wages of the workers at Universal Manufacturing of Melonville. Construct a histogram to represent the data (5 points).

x =hourly wage	Number of employees
$4.00 \leq x < 5.50$	21
$5.50 \leq x < 7.00$	35
$7.00 \leq x < 8.50$	40
$8.50 \leq x < 10.00$	27
$10.00 \leq x < 11.50$	19
$11.50 \leq x < 13.00$	10

23. A group of 10 students reported the number of hours that they studied for their Math 107 exam. The results are as follows.

0 1 3 5 5 7 11 12 13 15

- Find the mean of the data set (5 points).
- Find the median of the data set (3 points).
- Find the mode of the data set (3 points).
- Find the variance (5 points) and standard deviation (3 points) of the data set. Round your answers to 2 decimal places.

[illegible]

24. The amount of time required to assemble a component on a factory assembly line is normally distributed with a mean of 7.1 minutes and a standard deviation of 0.6 minute. Find the probability that a randomly selected employee will take between 6.7 and 7.6 minutes (8 points).

25. Shade the area under the bell curve for each probability (2 points). Circle the answer choice for the probability represented (2 points each).

(a) $p(-1.28 < z < 1.49)$

a) 0.3997

b) 0.4319

c) 0.8316

d) 0.0322

(b) $p(z > -1.23)$

a) 0.3907

b) 0.8907

c) 0.1093

d) 0.1271

26. An IQ test has a mean of 100 and a standard deviation of 15. Assuming that test scores are normally distributed, what percentage of the population is expected to score above 120? (5 points)

On my honor, I have neither given nor received any academic aid or information that would violate the Honor Code of Mars Hill University. _____(please sign)